NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-8805/98C(USAF)
31 MAR 1992

SUPERSEDING
MIL-S-8805/98B(USAF)
2 May 1978

PERFORMANCE SPECIFICATION SHEET

© SWITCHES, SENSITIVE SWITCH ASSEMBLY, TOGGLE, 7 AMPERES, PANEL SEAL, TOGGLE SEAL

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.

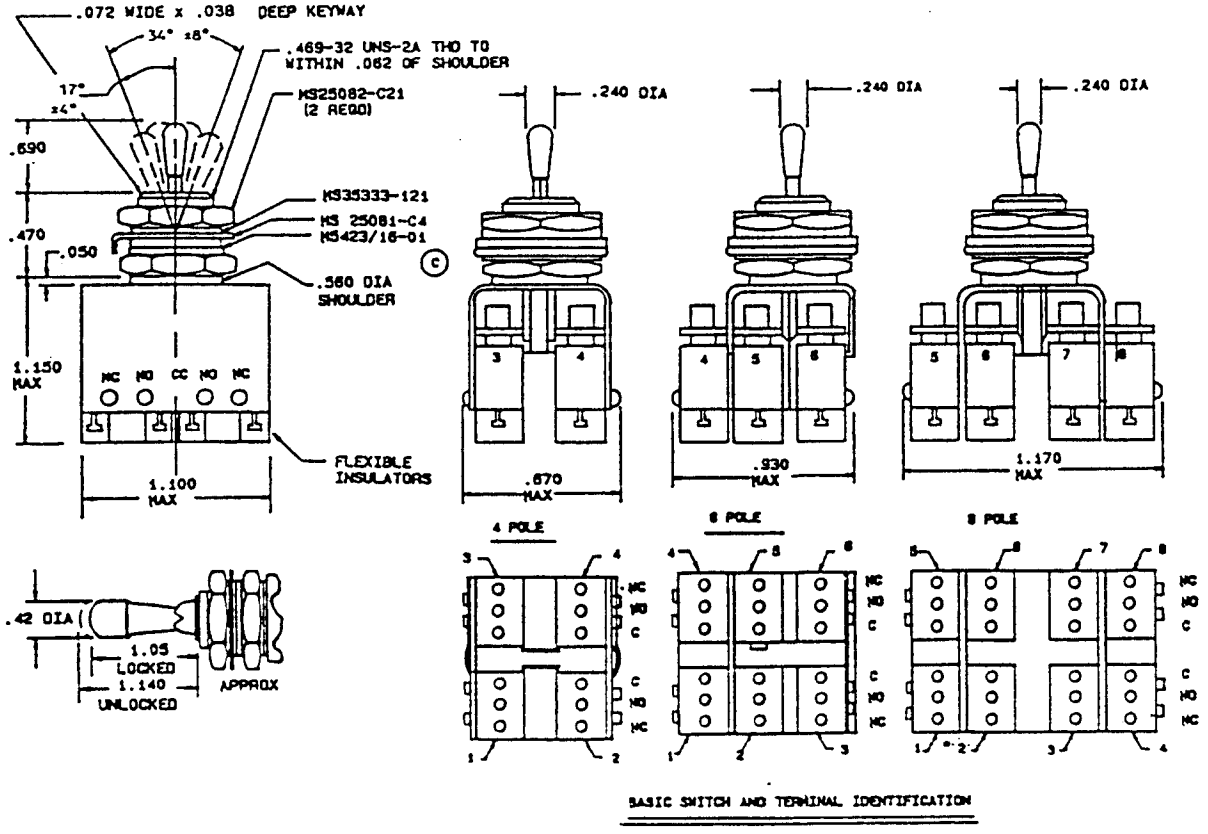


FIGURE 1. Configuration and dimensions.

© denotes changes

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Inches	mm	Inches	mm
038	0 97	.690	17.53
050	1.27	760	19 30
062	1 57	930	23 62
072	1 83	1 05	26 7
240	6.10	1.100	27 94
42	10.7	1 140	28 96
470	11.94	1.150	29 21
560	14.22	1.170	29 72
670	17 02	1 300	33 02

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ±.015 (0.38 mm) for decimals and ±5° for angles
4. Design configuration optional provided maximum dimensions are not exceeded
5. Basic switch identification shall be permanently marked on the basic switch as shown
6. Basic switch terminals shall be single turret (as shown) and shall accept AN-20 wire

FIGURE 1 Configuration and dimensions - Continued

(KEYWAY SIDE)

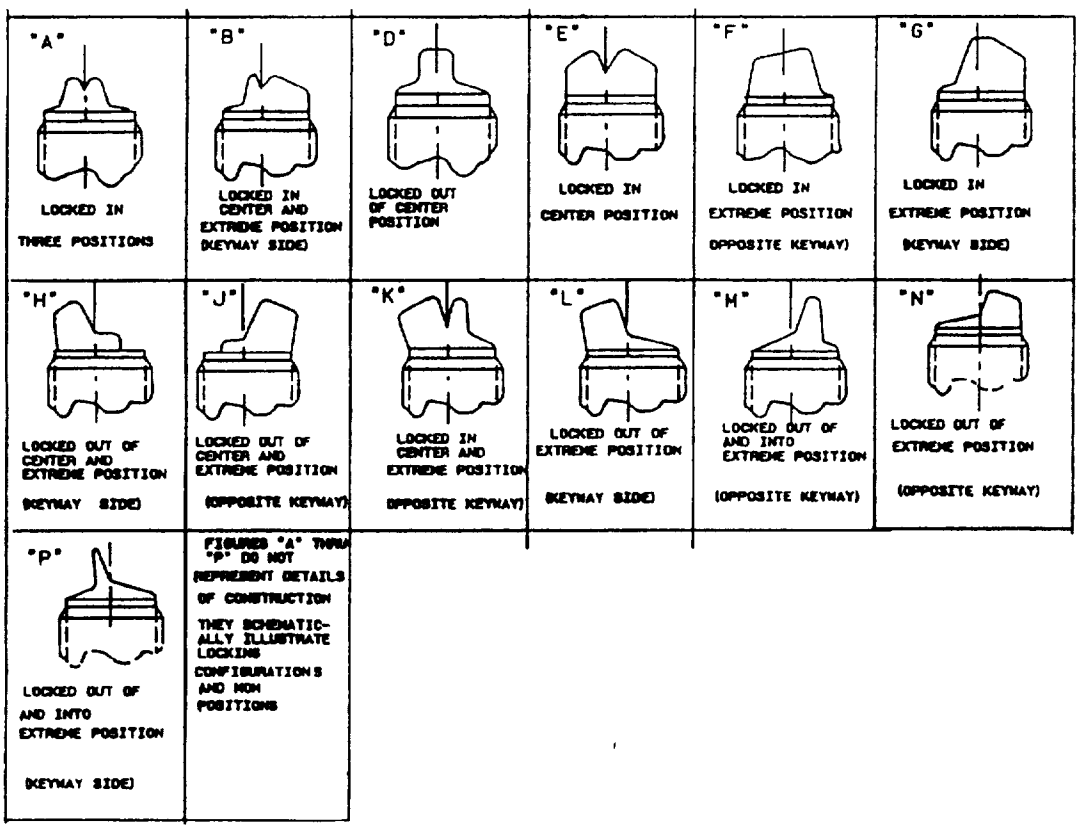


FIGURE 2 Locking arrangements

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REQUIREMENTS

Dimension and configuration: See figures 1 and 2

Enclosure design: 1 (unsealed) Switches shall have a molded silicone rubber toggle to bushing seal and a panel seal.

Temperature characteristics 1 (-55° to +85°C).

Shock type: 50 G, sawtooth, test condition G, method 213 of MIL-STD-202.

Vibration grade: Random vibration, tested in accordance with method 214 of MIL-STD-202 as follows:

20 - 300 Hz, 3 dB/octave increase
300 - 1000 Hz, 0.02 G^2/Hz
1000 - 2000 Hz, 6 dB/octave decrease

Weight (pounds maximum).

4 pole switches. .095 without lever lock, .110 with lever lock.
6 pole switches. .100 without lever lock, .115 with lever lock
8 pole switches: 110 without lever lock, 125 with lever lock.

Toggle lever The toggle lever shall be metal with a lusterless finish and shall have a fixed pivot

Locking arrangement The locking combinations for the lever-lock toggle lever shall provide positive locking and shall not permit a change in toggle lever position until the locking mechanism is manually released

Panel seal There shall be no leakage of water through the seals following the toggle seal test of MIL-S-83731

Dielectric withstanding voltage

Sea level 1,000 V rms
Altitude: 400 V rms at 50,000 feet

Terminal strength. 5 pounds

Strength of actuating means 25 pounds

Strength of mounting bushing 25 pound-inches shall be applied to the mounting nut

Mechanical endurance 50,000 cycles. Lever-lock switches shall also be tested for 50,000 cycles by pulling the lever to its fully extended position and returning to its fully locked position without causing contact transfer at a rate of 60 cycles per minute maximum

Electrical endurance 25,000 cycles

Electrical ratings: See table I

TABLE I. Electrical ratings

Load	Sea level		50,000 ft
	28 V dc	115 V ac, 60 Hz	28 V dc
	(Amperes)	(Amperes)	(Amperes)
Resistive	7	7	4
Inductive	4	7	2.5

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©QUALITY ASSURANCE.

The following tables for quality assurance inspections shall apply only when the basic sensitive switch is approved under specification sheet MIL-S-8805/4

Qualification inspection In accordance with table II.

Group A inspection In accordance with table III, provided in-process dielectric withstanding voltage is performed

TABLE II Qualification group submission inspection

Examination or test	Test sample	Other switch samples	Extent of approval
Group 1	Group 1	M8805/98-33 (6 units)	All Part or Identifying Numbers (PIN)
Visual and mechanical examination	M8805/98-15 (2 units)	1. Visual and mechanical examination	
Dielectric withstanding voltage	M8805/98-63 (4 units)	2. Operating characteristics	
Insulation resistance	M8805/98-81 (6 units)	3 Lever-lock mechanical endurance (4 units)	
Operating characteristics		4. Salt spray (corrosion) (2 units)	
Group 2	Group 2	M8805/98-64 (4 units)	
Strength of actuating means 1/	M8805/98-63 (4 units)	1 Strength of actuating means 1/	
Strength of mounting bushing 1/		2 Vibration	
Thermal shock		3 Shock (50 G)	
Vibration (random)		4 Moisture resistance	
Shock (50 G)		5 Operating characteristics	
Moisture resistance		6 Panel seal	
Dielectric withstanding voltage			
Operating characteristics			
Panel seal			
Group 3	Group 3		
Salt spray (corrosion)	M8805/98-81 (2 units)		
Group 4	Group 4		
Low temperature operation 1/	M8805/98-81 (4 units)		
Mechanical endurance-low temperature 2/			
Mechanical endurance-high temperature 1/			
Dielectric withstanding voltage			
Operating characteristics			
Panel seal			
Group 5	Group 5		
Electrical endurance	M8805/98-15 (2 units)		
Inductive load dc			

1/ Two sample units only For low temperature operation, lever shall be held in momentary position
2/ Same two sample units as for low temperature operation

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© TABLE III Group A inspection

Examination or test
Visual and mechanical examination
Operating characteristics

©PIN. M8805/98 - (dash number from table IV).

TABLE IV Dash number and characteristics

Dash number	Number of poles	Lever action and circuitry (table V)	Locking arrangement (figure 2)
-001	4 pole	1	None
-002			A
-003			B
-004			D
-005			E
-006			F
-007			G
-008			H
-009			J
-010			K
-011			L
-012			M
-013			N
-014			P
-015		3	None
-016			D
-017			F
-018			G
-019		5	None
-020			E
-021			F
-022			K
-023			L
-024			M
-025			N
-026		7	None
-027			E
-028			L
-029			N
-030		8	None
-031			F
-032	6 pole	1	None
-033			A
-034			B
-035			D
-036			E
-037			F
-038			G
-039			H
-040			J
-041			K
-042			L
-043			M
-044			N
-045			P

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TABLE IV Dash number and characteristics - Continued.

Dash number	Number of poles	Lever action and circuitry (table V)	Locking arrangement (figure 2)
-046	6 pole	3	None
-047			D
-048			F
-049			G
-050		5	None
-051			E
-052			F
-053			K
-054			L
-055			M
-056			N
-057		7	None
-058			E
-059			L
-060			N
-061		8	None
-062			F
-063	8 pole	1	None
-064			A
-065			B
-066			D
-067			E
-068			F
-069			G
-070			H
-071			J
-072			K
-073			L
-074			M
-075			N
-076			P
-077		3	None
-078			D
-079			F
-080			G
-081		5	None
-082			E
-083			F
-084			K
-085			L
-086			M
-087			N
-088		7	None
-089			E
-090			L
-091			N
-092		8	None
-093			F
-094 1/	4 pole	9	P

1/ Inactive for new design

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TABLE V Lever action and circuitry

Type number	Basic switch numbers			With toggle lever in position					
				Keying side		Center		Opposite keying side	
	For 4 pole	For 6 pole	For 8 pole	Toggle action	Circuit closed	Toggle action	Circuit closed	Toggle action	Circuit closed
1	1-2	1-2-3	1-2-3-4	Maintain	C-NC	Maintain	C-NC	Maintain	C-NO
	3-4	4-5-6	5-6-7-8		C-NO		C-NC		C-NC
3	1-2	1-2-3	1-2-3-4	Maintain	C-NC	(None)		Maintain	C-NO
	3-4	4-5-6	5-6-7-8		C-NO				C-NC
5	1-2	1-2-3	1-2-3-4	Momentary	C-NC	Maintain	C-NC	Maintain	C-NO
	3-4	4-5-6	5-6-7-8		C-NO		C-NC		C-NC
7	1-2	1-2-3	1-2-3-4	Momentary	C-NC	Maintain	C-NC	Momentary	C-NO
	3-4	4-5-6	5-6-7-8		C-NO		C-NC		C-NC
8	1-2	1-2-3	1-2-3-4	Momentary	C-NC	(None)		Maintain	C-NO
	3-4	4-5-6	5-6-7-8		C-NO				C-NC
9 1/	1-2	N/A	N/A	Momentary	C-NC	Momentary	C-NO	Maintain	C-NC
	3-4				C-NO		C-NC		C-NC

1/ Inactive for new design

CONCLUDING MATERIAL

Custodian
Air Force - 85

Review activity
Air Force - 99

Preparing activity
Air Force - 85

Agent.
DLA - ES

(Project 5930-F711)

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